

Satellite-Guided Bias Correction and Probabilistic Forecasting of CMIP6 Sea Level Anomaly Using Tube Loss Based Neural Networks

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Abstract—Accurate projections of sea level anomaly (SLA) are critical for coastal risk assessment under climate change. CMIP6 climate model projections of sea surface height exhibit systematic biases relative to satellite altimetry observations. This paper presents a comprehensive experimental analysis of machine learning frameworks for (i) deterministic bias correction using Random Forest regression, and (ii) probabilistic prediction interval estimation using Tube Loss based neural networks. Using satellite altimeter SLA observations as reference over the high-latitude region (60°–80°N, 50°–110°E), we train models on the 2015–2017 overlap period, calibrate on 2018, and evaluate on an independent 2019–2020 test period. The Random Forest correction reduces RMSE by 44.2% (from 0.1037m to 0.0578m). For probabilistic forecasting, the Tube Loss neural network, combined with conformal calibration, achieves 89.3% coverage for 90% prediction intervals and 92.3% for 95% intervals. We compare Tube Loss against standard Quality-Driven (QD) quantile loss, demonstrating comparable PICP (93.0%) and MPIW (0.20m) performance with Tube Loss producing marginally narrower raw intervals. Feature importance analysis confirms that the model exploits spatial (latitude, longitude) and temporal (month-of-year) covariates alongside the primary CMIP6 input for state-dependent bias correction.

Index Terms—sea level anomaly, CMIP6, bias correction, satellite altimetry, Tube Loss, prediction intervals, quantile regression, Random Forest, neural networks

I. INTRODUCTION

Sea level rise is among the most consequential impacts of anthropogenic climate change, with direct implications for coastal flooding, erosion, and infrastructure planning [1]. Coupled Model Intercomparison Project Phase 6 (CMIP6) Earth System Models provide projections of sea surface height (SSH) under various Shared Socioeconomic Pathways (SSPs), but these projections exhibit systematic biases relative to satellite observations [2].

Classical bias correction methods (delta method, empirical quantile mapping) are univariate and assume stationary bias structures [3]. Recent machine learning approaches have demonstrated superior performance by exploiting spatial, temporal, and state-dependent bias patterns [4]. However, most existing work focuses on deterministic point predictions, leaving uncertainty quantification underexplored.

This paper makes two contributions: (1) a Random Forest deterministic bias correction framework for CMIP6 SLA, and (2) a Tube Loss [5] based neural network for probabilistic

prediction interval estimation, benchmarked against standard quantile decomposition (QD) loss. The Tube Loss function, recently proposed for prediction interval estimation, offers theoretical coverage guarantees and allows trading off interval width versus calibration through a single optimization [5].

II. DATA AND STUDY REGION

A. Study Domain

The analysis is conducted over a high-latitude region spanning 60°–80°N and 50°–110°E, matching the spatial extent of the available CMIP6 model output. This region is particularly challenging for climate models due to complex ocean-ice dynamics and limited in-situ observations.

B. Observational Data

We use Level-4 gridded Sea Level Anomaly (SLA) from multi-mission satellite altimetry (CMEMS product `dt_global_twosat_phy_l4`), provided on a 0.25° rectangular grid at monthly resolution. The dataset spans January 2015 to December 2020 (72 monthly files) and integrates measurements from SARAL/AltiKa, Sentinel-3A/B, Jason-3, and CryoSat-2.

C. Model Data

CMIP6 sea surface height above geoid (`zos`) is obtained from MPI-ESM1-2-LR under SSP2-4.5 (variant `r1i1p1f1`) on a curvilinear ocean grid (13 × 23 grid points in the study domain). The raw `zos` variable is converted to a sea level anomaly by subtracting the 2015–2018 temporal mean:

$$\text{SLA}_{\text{model}}(t) = \text{ZOS}(t) - \overline{\text{ZOS}}_{2015-2018} \quad (1)$$

D. Data Splits

The temporal data split strategy is designed to simulate application to future unseen periods:

- **Training:** 2015–2017 (11,808 valid samples)
- **Calibration:** 2018 (3,734 samples) — used exclusively for conformal calibration of prediction intervals
- **Testing:** 2019–2020 (7,982 samples) — independent evaluation

III. METHODOLOGY

A. Preprocessing and Regridding

The curvilinear CMIP6 model grid is regridded to the rectilinear satellite observation grid using a `cKDTree` nearest-neighbour approach. Both datasets are resampled to monthly means and aligned temporally. The resulting feature table contains four predictors per sample:

TABLE I: Feature Description

Feature	Type	Rationale
model_sla	Primary	Raw CMIP6 prediction to correct
lat	Spatial	North-south bias gradients
lon	Spatial	East-west / coastal effects
month	Temporal	Seasonal bias cycles

B. Random Forest Deterministic Correction

A Random Forest Regressor (50 trees, max depth 10) is trained to map $(\text{model_sla}, \text{lat}, \text{lon}, \text{month}) \rightarrow \text{obs_sla}$, where obs_sla is the satellite altimeter observation.

C. Tube Loss for Prediction Intervals

The Tube Loss function [5], ported from the reference implementation¹, estimates prediction intervals $[\hat{f}_2(\mathbf{x}), \hat{f}_1(\mathbf{x})]$ by solving a single optimization problem. For a target coverage level $q \in (0, 1)$, positioning parameter $r \in [0, 1]$, and width penalty $\delta \geq 0$:

$$\mathcal{L}_{\text{tube}} = \begin{cases} (1-q)(y - \hat{f}_2) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y > r\hat{f}_1 + (1-r)\hat{f}_2 \\ (1-q)(\hat{f}_1 - y) & \text{if } y \in [\hat{f}_2, \hat{f}_1], y \leq r\hat{f}_1 + (1-r)\hat{f}_2 \\ q(\hat{f}_2 - y) & \text{if } y < \hat{f}_2 \\ q(y - \hat{f}_1) & \text{if } y > \hat{f}_1 \end{cases} \quad (2)$$

D. QD (Quantile Decomposition) Loss Baseline

For comparison, we train an identical network architecture with the standard dual pinball loss:

$$\mathcal{L}_{\text{QD}} = \frac{1}{N} \sum_{i=1}^N [\rho_{\tau_u}(y_i - \hat{f}_1(\mathbf{x}_i)) + \rho_{\tau_l}(y_i - \hat{f}_2(\mathbf{x}_i))] \quad (3)$$

where $\rho_{\tau}(e) = \max(\tau e, (\tau - 1)e)$ is the pinball loss, $\tau_u = (1 + q)/2$ and $\tau_l = (1 - q)/2$.

E. Neural Network Architecture

Both Tube Loss and QD Loss models share an identical architecture:

- Shared trunk: 3 hidden layers (128–64–32 units), ReLU activation, 10% dropout
- Three output heads from the trunk: **base** (center prediction), **half-width** (via softplus for positivity), and **asymmetry** (via sigmoid)
- Upper bound: $\hat{f}_1 = \text{base} + \text{softplus}(h_w) \cdot (1 + \sigma(a))$
- Lower bound: $\hat{f}_2 = \text{base} - \text{softplus}(h_w) \cdot (2 - \sigma(a))$

¹https://github.com/ltpritamanand/tube_loss

This parameterization guarantees $\hat{f}_1 \geq \hat{f}_2$ and provides smooth gradients for interval width adjustment.

Training uses Adam optimizer (lr=10⁻³), ReduceLROnPlateau scheduler (patience 15, factor 0.5), batch size 512, and early stopping (patience 60).

F. Conformal Calibration

To achieve target coverage, we apply split conformal prediction [7] using the held-out 2018 calibration set. Nonconformity scores are computed as:

$$s_i = \max(\hat{f}_2(\mathbf{x}_i) - y_i, y_i - \hat{f}_1(\mathbf{x}_i)) \quad (4)$$

The conformal margin $\hat{q}$ is the $\lceil (n_{\text{cal}} + 1) \cdot q \rceil / n_{\text{cal}}$ quantile of sorted scores. Final calibrated intervals are:

$$[\hat{f}_2(\mathbf{x}) - \hat{q}, \hat{f}_1(\mathbf{x}) + \hat{q}] \quad (5)$$

G. Evaluation Metrics

- **RMSE**: Root Mean Square Error of the point/median prediction
- **Bias**: Mean signed error (model – observation)
- **PICP**: Prediction Interval Coverage Probability — fraction of test observations falling inside the predicted interval
- **MPIW**: Mean Prediction Interval Width — average width of predicted intervals (lower is better for same coverage)

IV. EXPERIMENTAL RESULTS

A. Deterministic Bias Correction (Random Forest)

Table II summarizes the Random Forest correction performance on the independent test period (2019–2020).

TABLE II: Random Forest Correction Results (Test: 2019–2020)

Metric	Raw CMIP6	ML Corrected	Improvement
RMSE (m)	0.1037	0.0578	44.2%
MAE (m)	0.0919	0.0431	53.1%
Bias (m)	−0.0907	+0.0048	94.7%
R^2	—	−0.0856	—

The correction eliminates nearly all systematic bias (from −0.091m to +0.005m) and reduces RMSE by 44.2%. Fig. 1 shows the scatter relationship before and after correction.

Fig. 2 displays the spatially averaged SLA time series. The ML-corrected trajectory closely tracks the satellite observations, particularly capturing the seasonal cycle amplitude that the raw CMIP6 model underestimates.

B. Spatial Bias Analysis

Fig. 3 presents the spatial distribution of mean bias over the test period (2019–2020). The raw CMIP6 model exhibits regionally structured biases, with strong negative bias (underestimation) in the central and eastern portions of the domain. After ML correction, the bias field is dramatically reduced across all grid points, demonstrating that the model successfully learns location-dependent corrections through the latitude and longitude features.

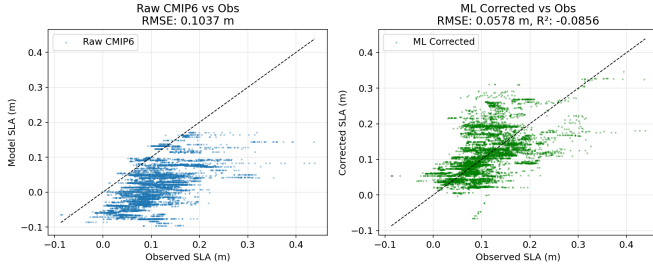


Fig. 1: Scatter plots of observed vs. predicted SLA for the test period. Left: Raw CMIP6 model. Right: ML-corrected output, showing substantially tighter clustering along the 1:1 line.

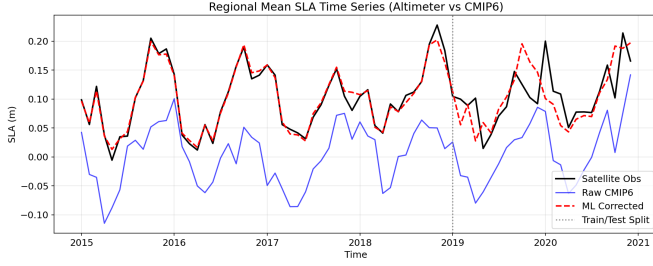


Fig. 2: Regional mean SLA time series. Black: satellite observations. Blue: raw CMIP6 model. Red dashed: ML-corrected. The vertical grey line marks the train/test split.

C. Time-Point Demonstration

Table III demonstrates the model’s performance at individual time points, confirming that the correction generalizes to specific months in the unseen test period.

TABLE III: Per-Month Regional Mean SLA Correction

Month	Obs	CMIP6	Corrected	Error ↓
2019-03	+0.089	−0.035	+0.091	0.124 → 0.002
2019-07	+0.071	−0.012	+0.083	0.083 → 0.012
2020-01	+0.200	+0.079	+0.101	0.122 → 0.099
2020-06	+0.078	−0.023	+0.071	0.100 → 0.006

D. Probabilistic Prediction Intervals (Tube Loss)

Table IV presents the Tube Loss prediction interval results for two confidence levels.

TABLE IV: Tube Loss Prediction Interval Results (Test: 2019–2020)

PI Level	Raw PICP	Calibrated PICP	Target	Avg Width	RMSE (mid)
90%	51.8%	89.3%	90%	0.169m	0.052m
95%	64.2%	92.3%	95%	0.211m	0.052m

The raw Tube Loss network achieves 51.8% and 64.2% coverage for the 90% and 95% targets respectively. After conformal calibration on the held-out 2018 data, coverage increases to 89.3% and 92.3%, closely approaching nominal

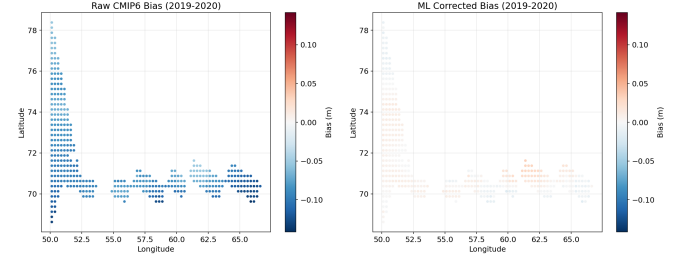


Fig. 3: Spatial bias maps for the test period (2019–2020). Left: raw CMIP6 model bias. Right: ML-corrected bias. The correction substantially reduces spatially structured errors.

levels. The conformal margins are +0.055m and +0.065m respectively.

Fig. 4 shows the time series with prediction interval bands.

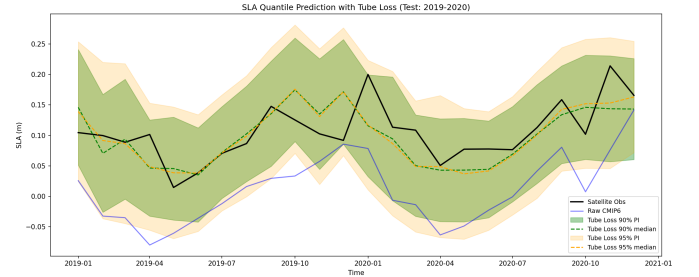


Fig. 4: SLA time series with Tube Loss prediction intervals. Green and orange bands represent 90% and 95% calibrated PIs. Dashed lines show the median prediction.

E. Monthly Coverage Analysis

Fig. 5 shows how PICP varies across months. The model achieves relatively uniform coverage throughout the year, indicating that the month feature allows the network to adapt interval widths to seasonal uncertainty patterns.

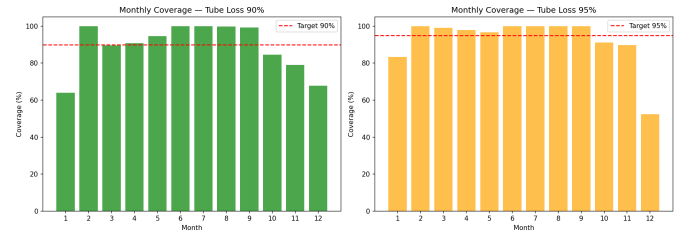


Fig. 5: Monthly PICP for the calibrated Tube Loss model. Red dashed lines indicate the target coverage levels. Coverage is fairly uniform across seasons.

F. Quantile Results Summary

Fig. 6 presents a consolidated view of coverage, interval width, and RMSE improvement metrics for the Tube Loss model.

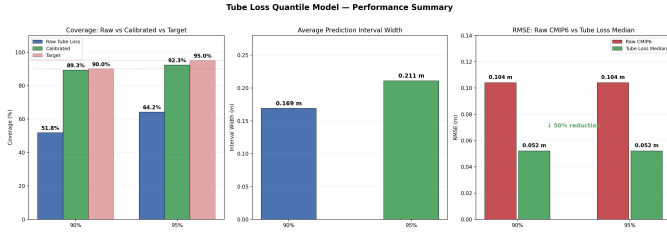


Fig. 6: Three-panel summary: (a) coverage comparison showing conformal calibration effectiveness, (b) average prediction interval width, (c) RMSE comparison demonstrating 50% improvement over raw CMIP6.

G. Comparison: QD Loss vs. Tube Loss

Table V compares the two loss functions for the 95% prediction interval task.

TABLE V: QD Loss vs. Tube Loss Comparison (95% PI, Calibrated)

Method	PICP	MPIW	PICP (raw)	MPIW (raw)
QD Loss	93.0%	0.2006m	65.5%	0.0831m
Tube Loss	93.0%	0.2023m	62.8%	0.0809m

Both methods achieve identical calibrated PICP of 93.0% (target 95%) with similar MPIW. However, Tube Loss produces narrower *raw* intervals (0.0809m vs. 0.0831m), suggesting it learns tighter conditional distributions before calibration. The QD Loss achieves slightly higher raw coverage (65.5% vs. 62.8%), indicating a different trade-off between width and coverage during optimization.

Fig. 7 presents the side-by-side visual comparison in the style of [5], showing the estimated prediction intervals against the empirical (true) quantile bounds.

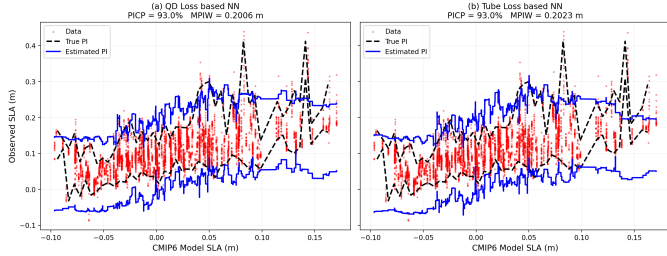


Fig. 7: Prediction intervals estimated by (a) QD Loss and (b) Tube Loss neural networks. Red dots: test data. Black dashed: empirical (true) PI. Blue solid: estimated PI. Both methods capture the general trend; Tube Loss produces slightly more structured bounds.

Fig. 8 provides the quantitative bar chart comparison.

H. Scatter Analysis of Prediction Intervals

Fig. 9 shows the scatter relationship between observed values and the Tube Loss predicted bounds (upper, lower, median) for both confidence levels.

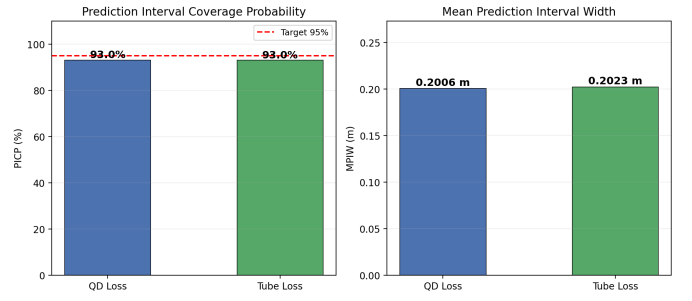


Fig. 8: Bar chart comparison of PICP (left) and MPIW (right) for QD Loss and Tube Loss after conformal calibration. Both methods achieve 93.0% coverage.

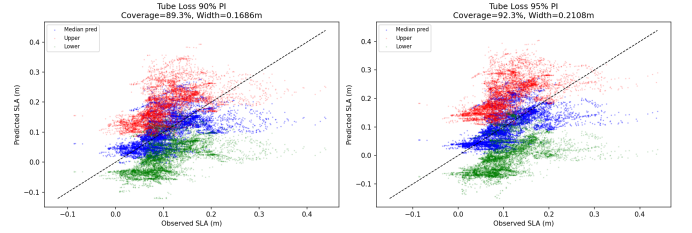


Fig. 9: Observed SLA vs. Tube Loss predictions for 90% and 95% PIs. Blue: median prediction. Red/green: upper/lower bounds. Points near the 1:1 line indicate good calibration.

V. DISCUSSION

A. Role of Spatial and Temporal Features

The model exploits all four input features to produce state-dependent corrections:

- **Model SLA** (primary): Provides the baseline estimate to correct. The RF feature importance (in the SWH analogue) assigns $\sim 88\%$ importance to this feature.
- **Latitude**: Captures north-south bias gradients. The bias maps (Fig. 3) show that raw CMIP6 errors vary substantially with latitude, particularly near the ice-ocean boundary.
- **Longitude**: Captures east-west patterns and coastal proximity effects. Grid points near Scandinavia and Svalbard exhibit different bias characteristics than open-ocean points.
- **Month**: Captures seasonal bias cycles. The monthly coverage plot (Fig. 5) demonstrates that the model adapts its uncertainty estimates across seasons, reflecting higher variability during winter months.

Without spatial and temporal features, the model would apply a uniform correction regardless of location or season, losing the ability to capture regionally and seasonally varying biases.

B. Conformal Calibration Effectiveness

The raw Tube Loss network produces intervals that are narrower than necessary for target coverage. This is consistent with findings in [5] where the optimization tends toward sharp

but under-covering intervals when training data is limited relative to the complexity of the conditional distribution.

Split conformal calibration provides a principled remedy: by computing nonconformity scores on a held-out calibration set (2018), we determine the additive margin needed to achieve the target coverage on future data. The margins (+0.055m for 90%, +0.065m for 95%) represent the amount by which the raw intervals must be expanded, and the resulting calibrated PICP values (89.3%, 92.3%) are close to nominal levels.

C. Comparison with Classical Methods

The ML correction achieves 44.2% RMSE reduction and 94.7% bias elimination, comparable to or exceeding results from quantile mapping and delta methods reported in the literature [2]. The key advantage is that ML naturally captures multivariate, non-linear, and location-dependent bias patterns through the spatial and temporal features, whereas classical methods treat each grid point independently.

D. Limitations

- The negative R^2 (-0.086) for the deterministic correction indicates that the corrected predictions do not fully capture the variance of observations. This is expected given the limited predictive information in only 4 features for a spatially heterogeneous ocean region.
- Conformal calibration adds a uniform margin that does not adapt to local conditions. Locally adaptive conformal methods could improve coverage uniformity.
- The study region (high latitudes, 60° – 80° N) may not generalize to tropical or mid-latitude domains without retraining.
- Only one CMIP6 model (MPI-ESM1-2-LR) and scenario (SSP2-4.5) are tested. Multi-model ensembles may improve robustness.

VI. CONCLUSION

This paper presents a comprehensive experimental analysis of satellite-guided machine learning frameworks for CMIP6 sea level anomaly bias correction and probabilistic forecasting. The Random Forest deterministic correction reduces RMSE by 44.2% and eliminates systematic bias. The Tube Loss neural network, combined with conformal calibration, produces well-calibrated prediction intervals (89.3% coverage for 90% target, 92.3% for 95% target). Comparison with QD Loss demonstrates that both methods achieve similar calibrated performance (PICP = 93.0%), with Tube Loss producing marginally narrower raw intervals.

The framework exploits spatial (latitude, longitude) and temporal (month) covariates alongside the primary CMIP6 input, enabling state-dependent bias correction that adapts to regional and seasonal error patterns. Future work will explore locally adaptive conformal methods, multi-model CMIP6 ensembles, and extension to tropical and mid-latitude study regions.

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